GREEEN tallgrass prairie restoration: Seeded richness ~ Species pool size (Lux Arbor)

MB

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NOTE: To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
```

```r
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    seeded_pool = col_factor(
      levels = c("0", "6", "12", "18", "33"), ordered = TRUE
      ),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 5.381342 4.878049 4.374756
```

```r
median(sites$y)
```

```
## [1] 4
```

```r
sd(sites$y)
```

```
## [1] 4.331835
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##  5% 95%
##   0  12
```

```r
sites %>% count(year)
```

```
## # A tibble: 6 x 2
##   year      n
##   <fct> <int>
## 1 2015     47
## 2 2016     48
## 3 2017     48
## 4 2018     48
## 5 2019     48
## 6 2025     48
```

```r
sites %>% count(seeded_pool)
```

```
## # A tibble: 4 x 2
##   seeded_pool     n
##   <ord>       <int>
## 1 6              72
## 2 12             72
## 3 18             72
## 4 33             71
```

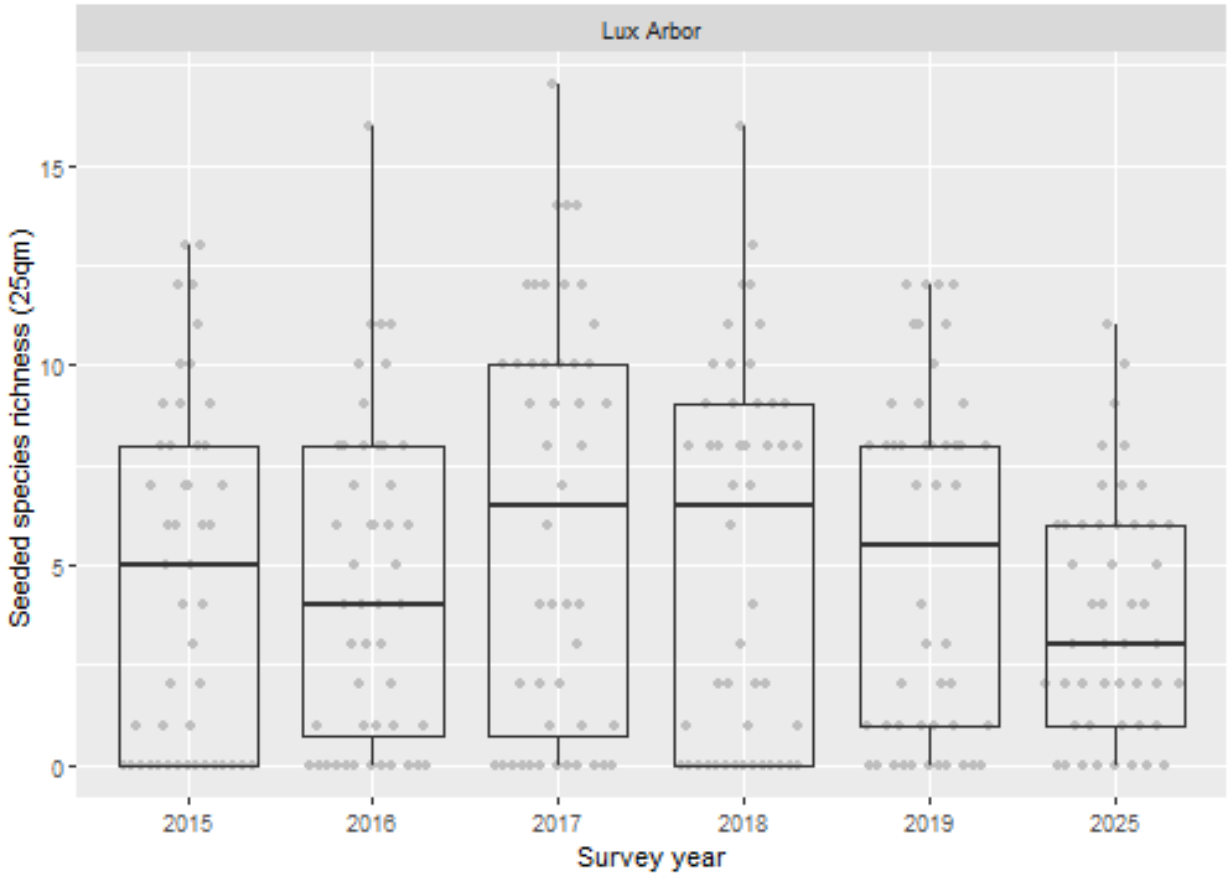
```r
sites %>% count(seeding_time)
```

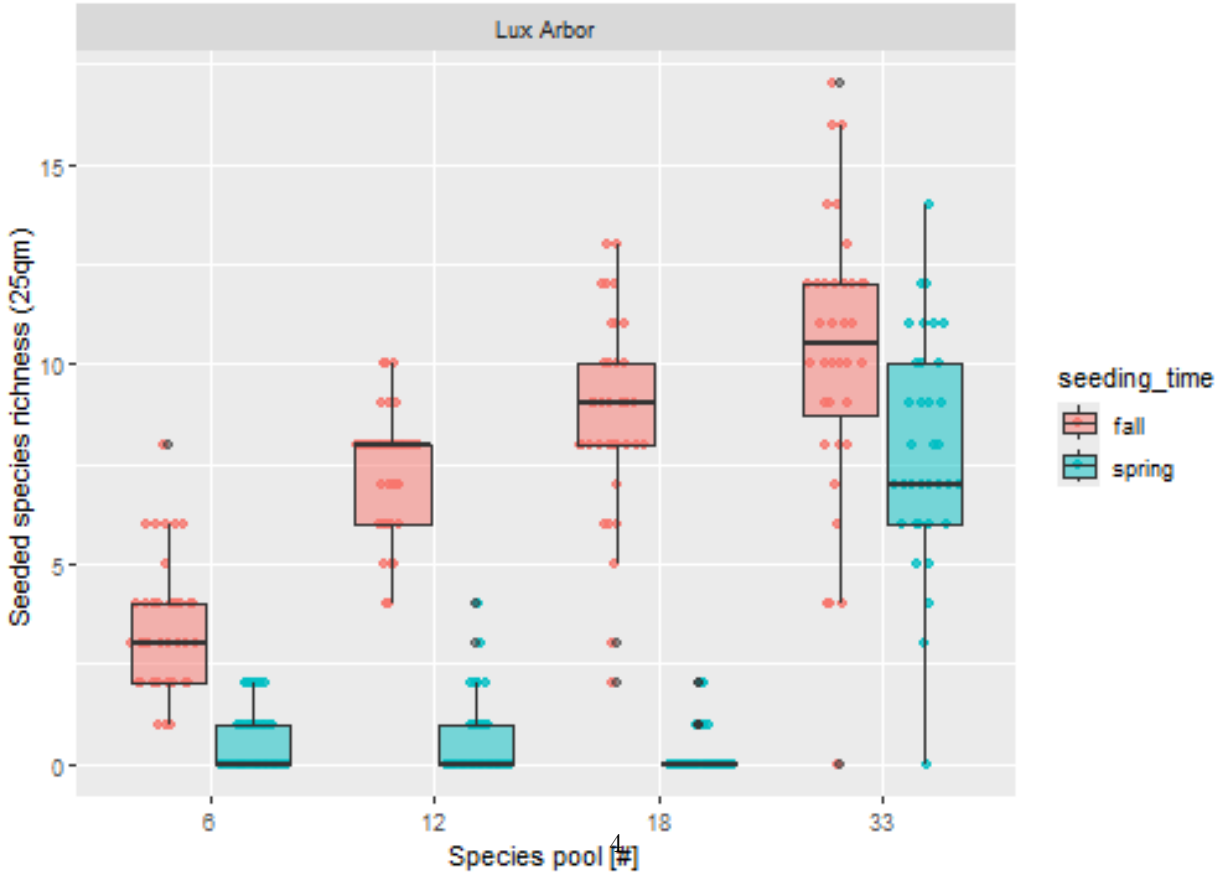
```
## # A tibble: 2 x 2
##   seeding_time     n
##   <fct>        <int>
## 1 fall           144
## 2 spring         143
```

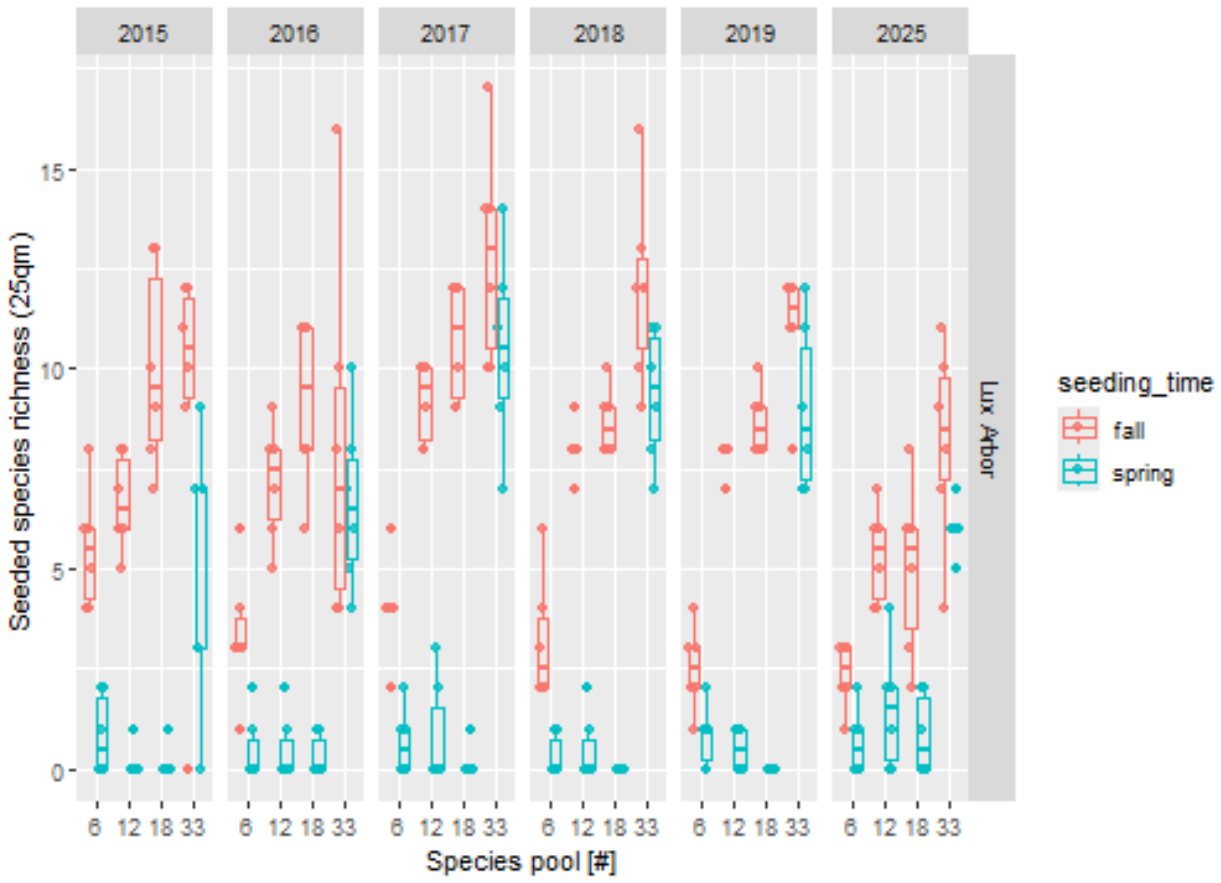
```r
sites %>% count(herbicide)
```

```
## # A tibble: 2 x 2
##   herbicide     n
##       <dbl> <int>
## 1         0   144
## 2         1   143
```

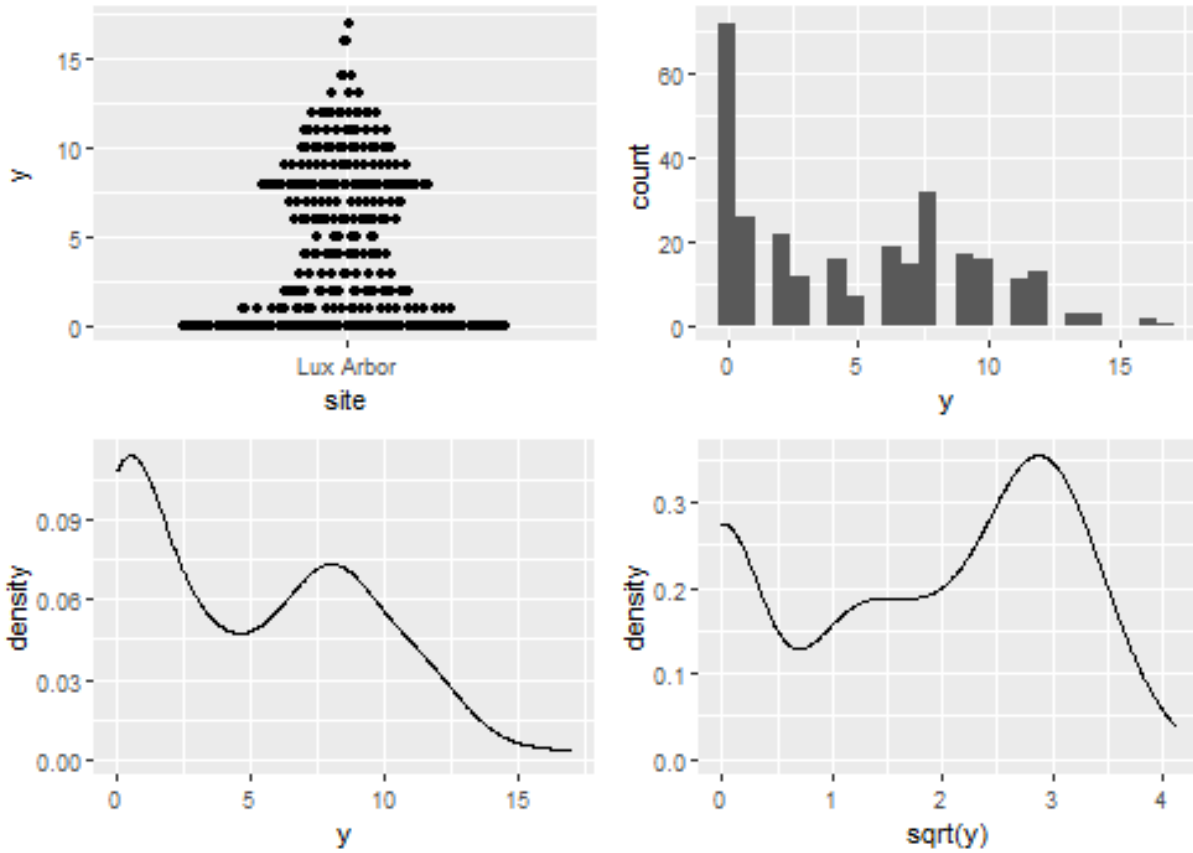
Graphs of raw data (Step 2, 6, 7)






2015
2016
2017
2018
2019
2025
Lux Arbor
Seeded species richness (25qm)
0
5
10
15
6 12 18 33
Species pool [#]
seeding_time
fall
spring

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_richness_pool_lux_2.Rdata"))
load(file = here("outputs", "models", "model_richness_pool_lux_3.Rdata"))
m_1 <- m2
m_2 <- m3
```

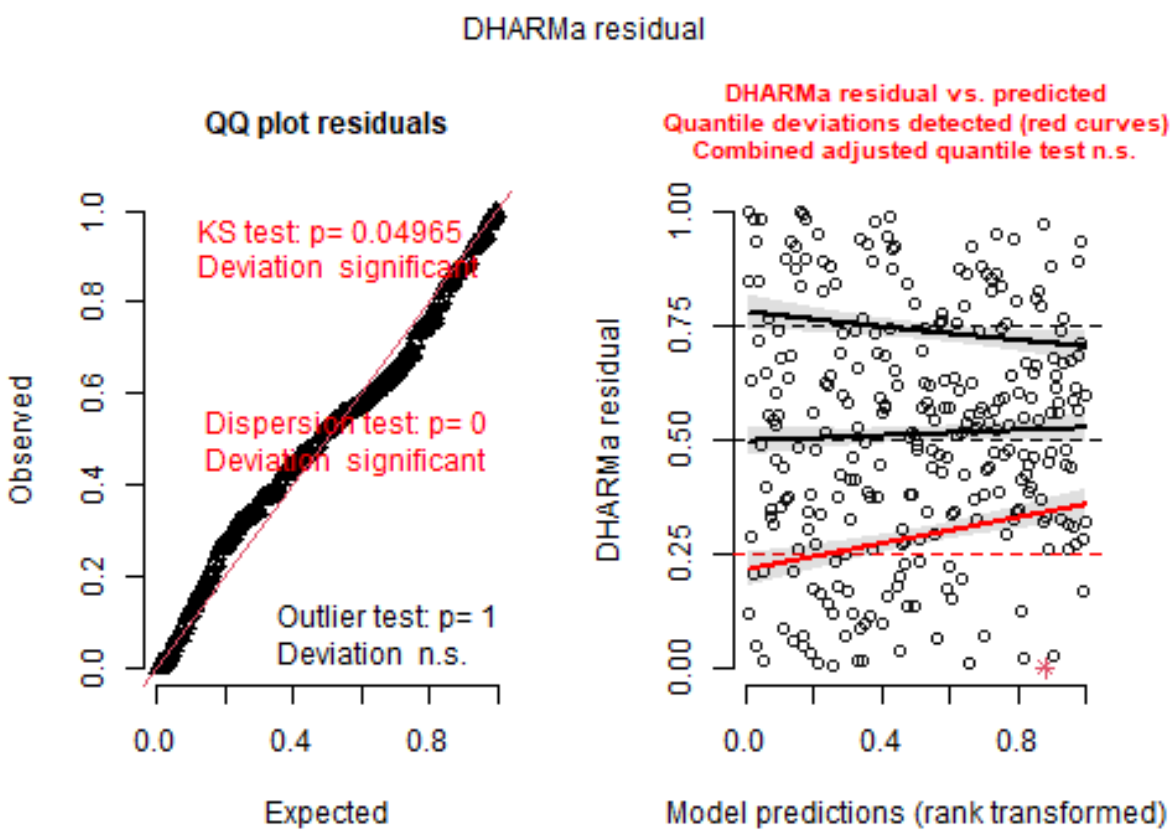
```
formula(m_1)
## y ~ seeded_pool * seeding_time + year + water_cap
formula(m_2)
```

```r
## y ~ seeded_pool * seeding_time + year + water_cap
# m_1@call
# m_2@call
class(m_1)
## [1] "glm" "lm"
class(m_2)
## [1] "glmmTMB"
```

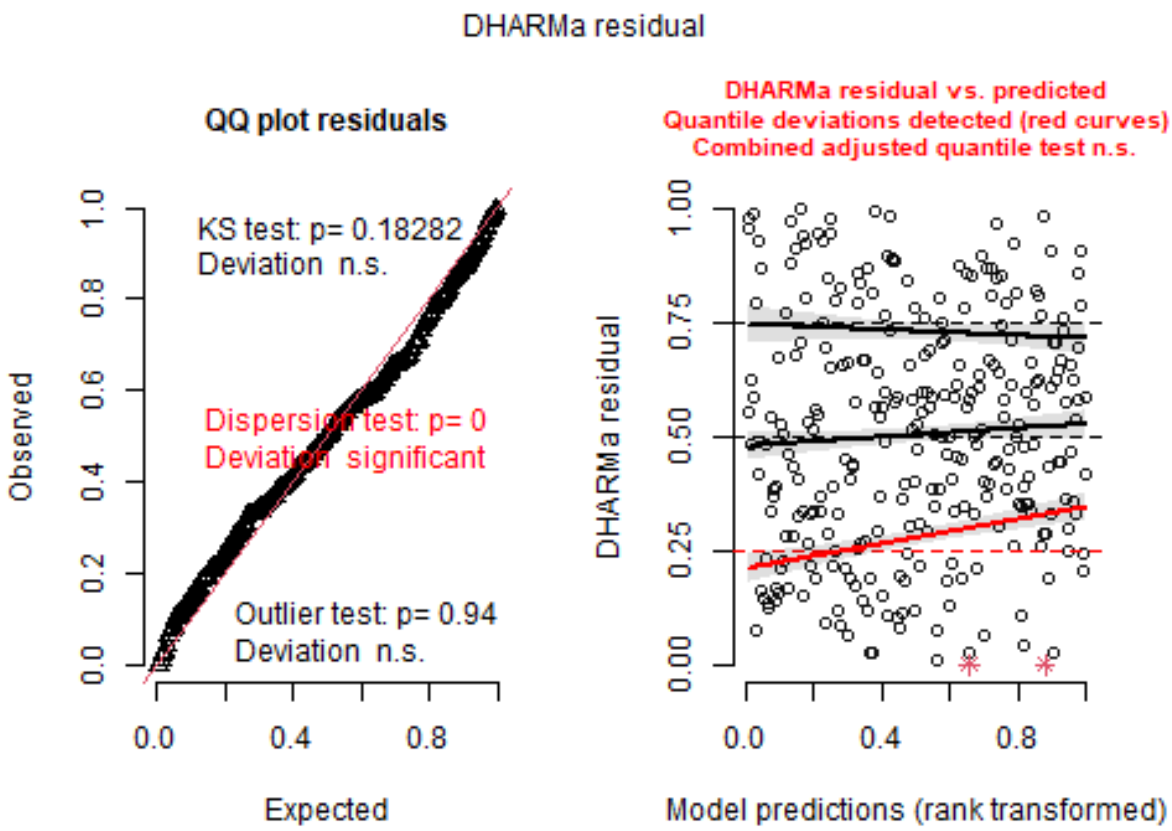
Model check

DHARMa

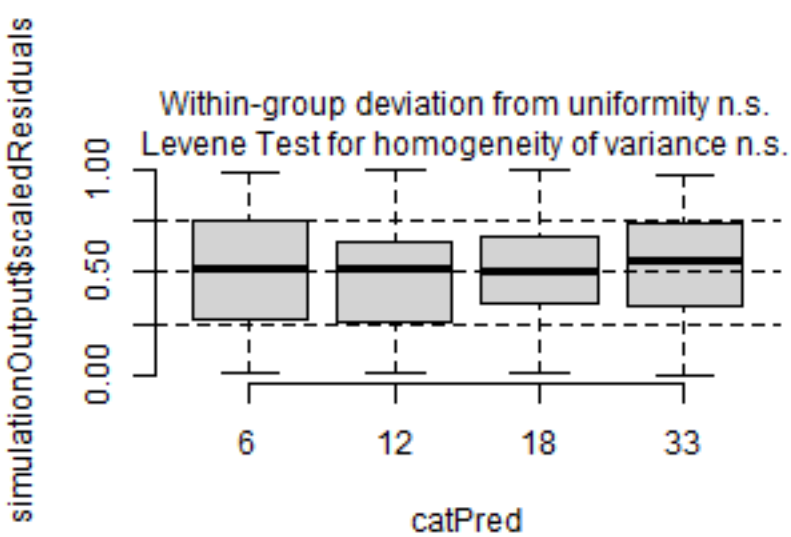
```r
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



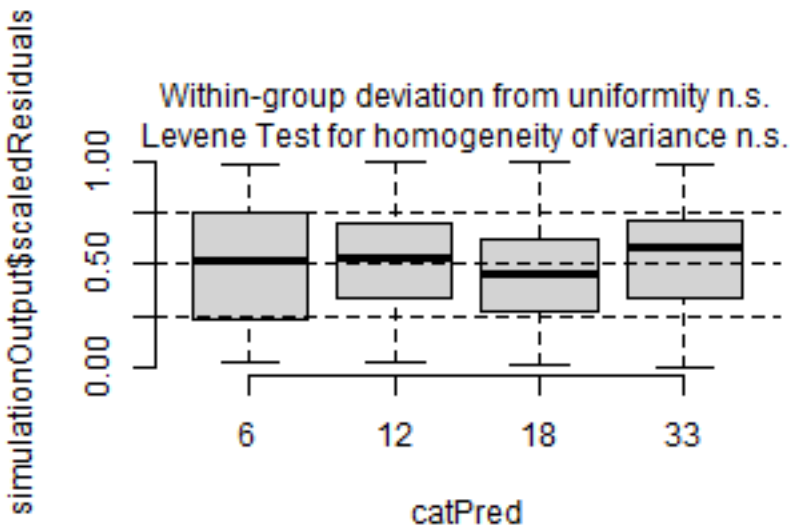

```r
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



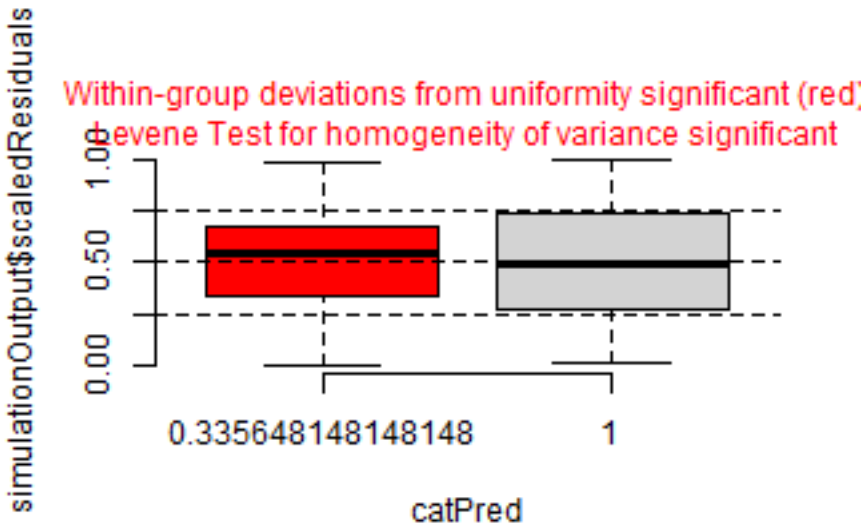

```
plotResiduals(simulation_output_1$scaledResiduals, factor(sites$seeded_pool))
```



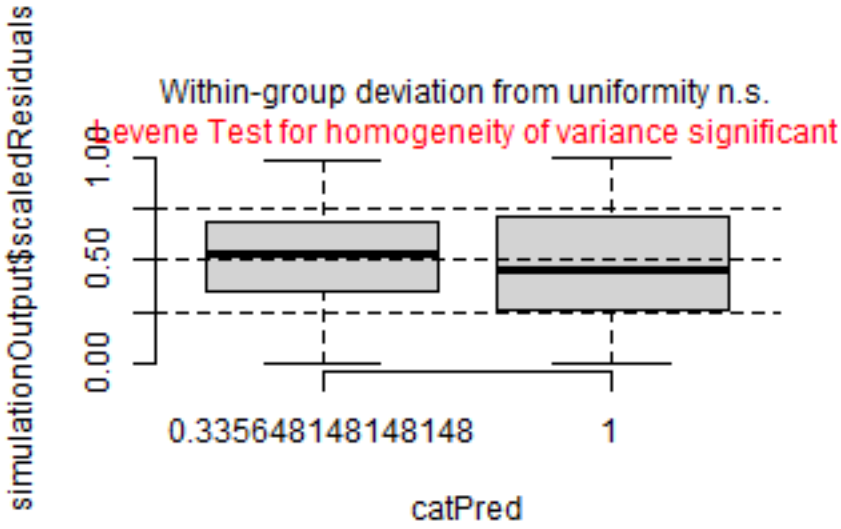

```
plotResiduals(simulation_output_2$scaledResiduals, factor(sites$seeded_pool))
```



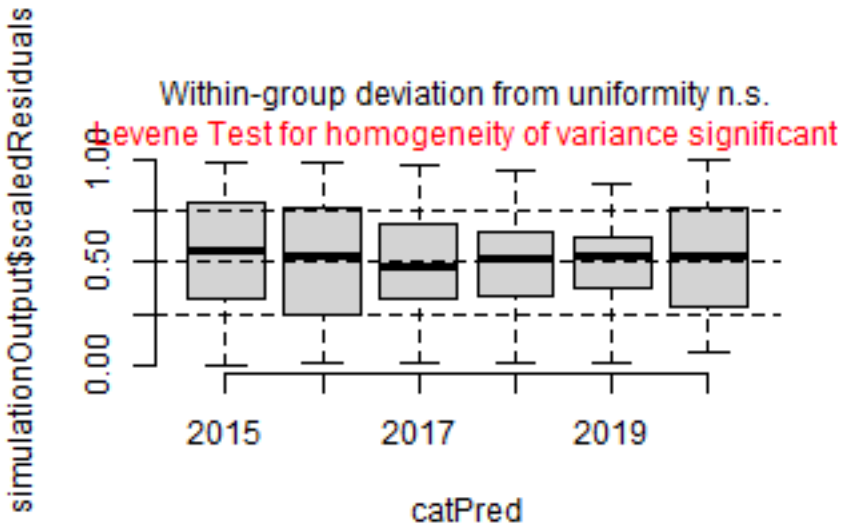

```
plotResiduals(simulation_output_1$scaledResiduals, sites$herbicide)
```



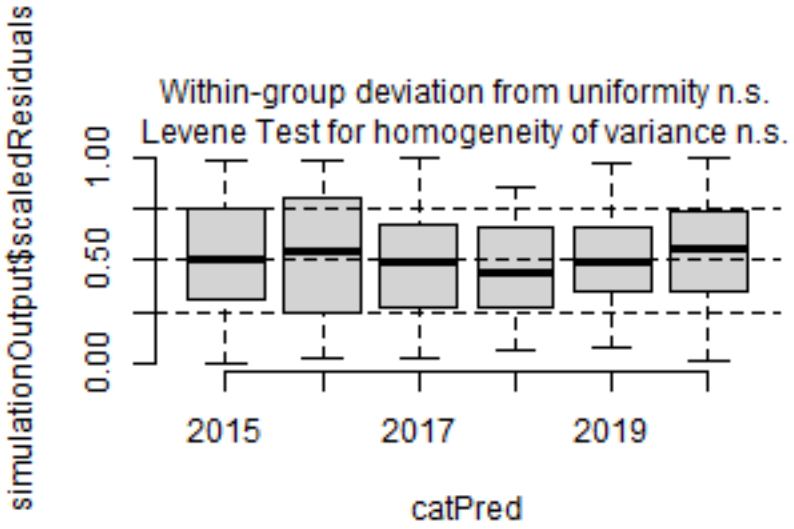

```
plotResiduals(simulation_output_2$scaledResiduals, sites$herbicide)
```



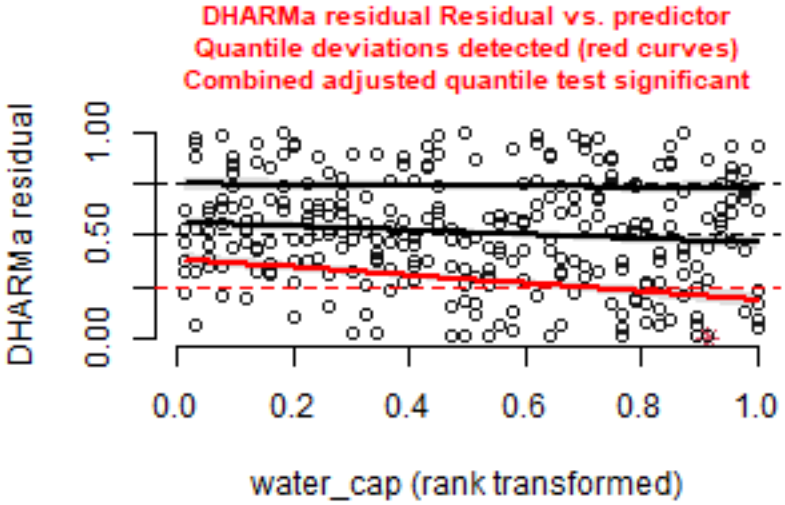

```
plotResiduals(simulation_output_1$scaledResiduals, sites$year)
```



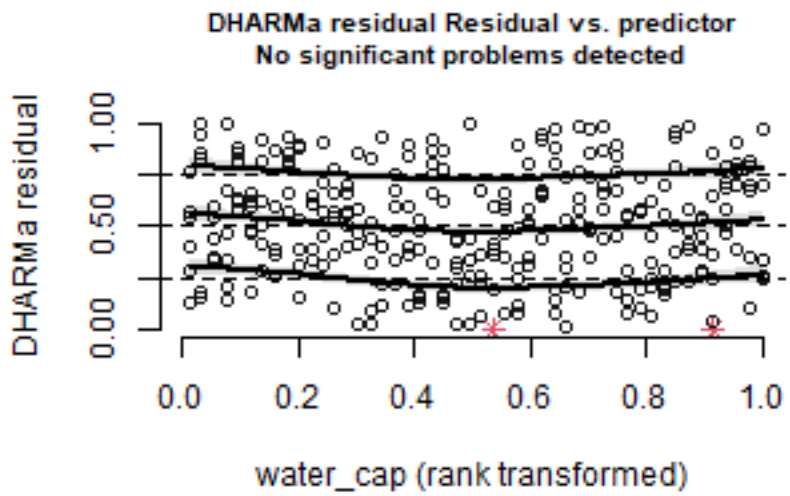

```
plotResiduals(simulation_output_2$scaledResiduals, sites$year)
```




```
plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
```

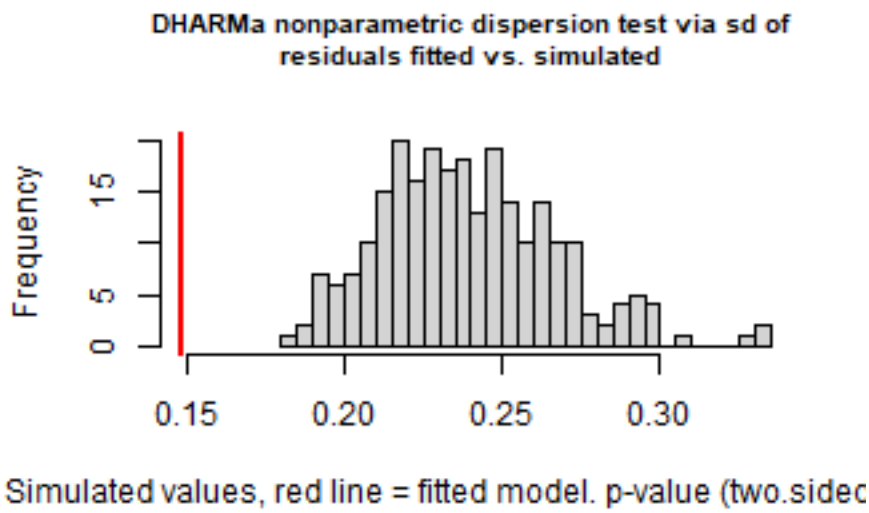



```
plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
```

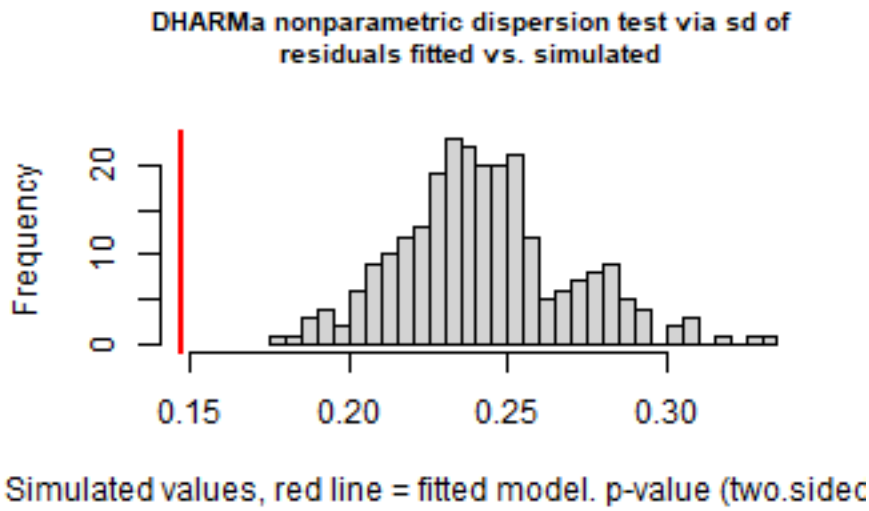


Test for overdispersion

```
testDispersion(simulation_output_1)
```




```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.6211, p-value < 2.2e-16
## alternative hypothesis: two.sided
testDispersion(simulation_output_2)
```




```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.60787, p-value < 2.2e-16
## alternative hypothesis: two.sided
```

Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
# car::vif(m_1)
# car::vif(m_2)
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                 R2m       R2c
## delta     0.9030481 0.9030481
## lognormal 0.9110277 0.9110277
## trigamma  0.8935664 0.8935664
MuMIn::r.squaredGLMM(m_2)
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                 R2m       R2c
## delta     0.9030481 0.9030481
## lognormal 0.9110278 0.9110278
## trigamma  0.8935664 0.8935664
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df     AICc
## m_1 14 1033.244
## m_2 15 1035.471
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##                              Chisq Df Pr(>Chisq)
## (Intercept)                63.1046  1  1.960e-15 ***
## seeded_pool               110.0165  3  < 2.2e-16 ***
## seeding_time              287.8923  1  < 2.2e-16 ***
## year                       30.9165  5  9.729e-06 ***
## water_cap                   1.6918  1     0.1934
## seeded_pool:seeding_time 152.9720  3  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

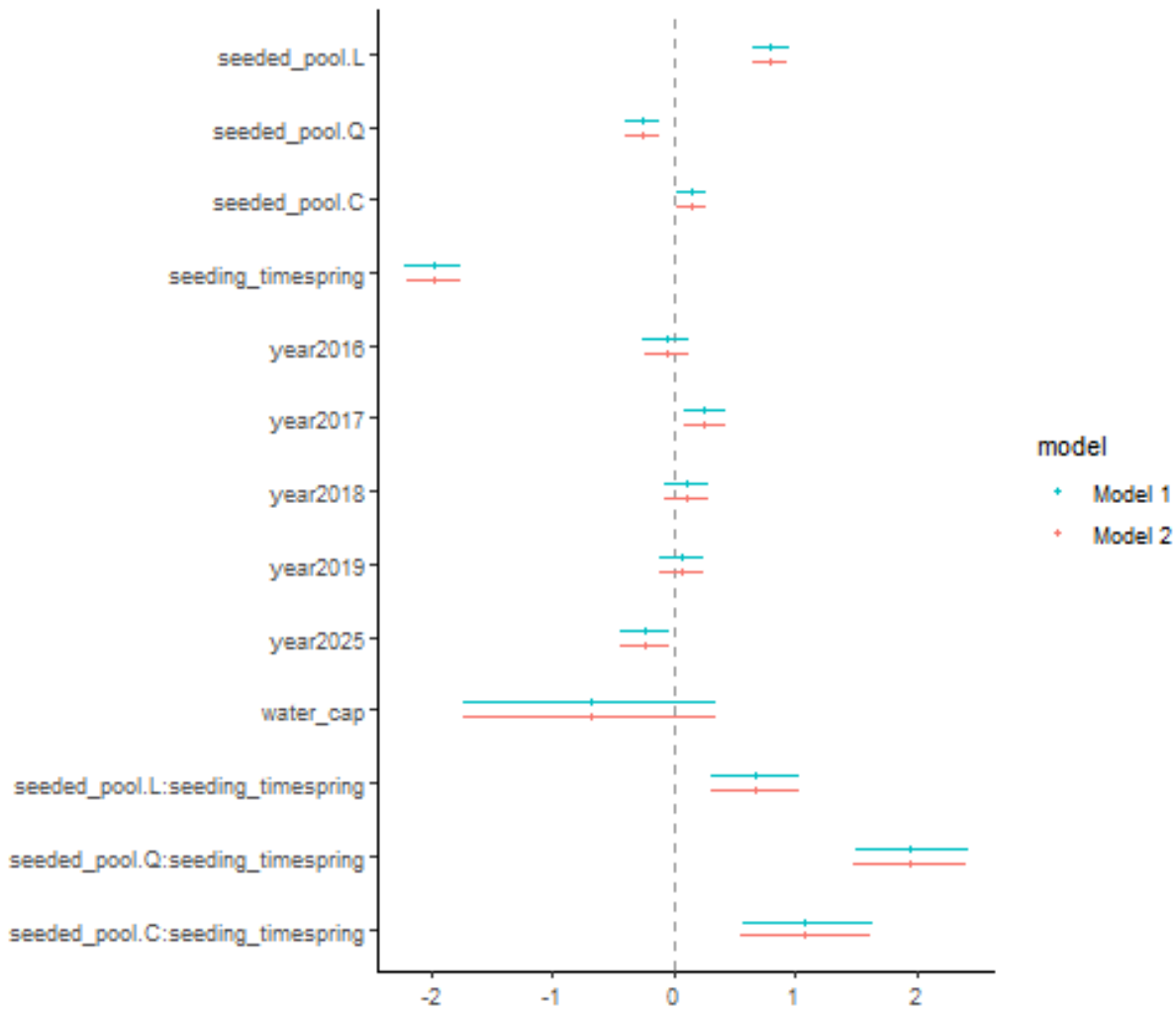
```
summary(m_2)
```

```
##  Family: nbinom2  ( log )
## Formula:          y ~ seeded_pool * seeding_time + year + water_cap
## Data: sites
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1033.7    1088.6    -501.9    1003.7       272
##
##
## Dispersion parameter for nbinom2 family (): 4.36e+08
##
## Conditional model:
##                                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)                       2.25974    0.28446   7.944 1.96e-15 ***
## seeded_pool.L                     0.79302    0.07676  10.331  < 2e-16 ***
## seeded_pool.Q                    -0.26570    0.07089  -3.748 0.000178 ***
## seeded_pool.C                     0.14498    0.06180   2.346 0.018971 *
## seeding_timespring               -1.97930    0.11665 -16.967  < 2e-16 ***
## year2016                         -0.06216    0.09603  -0.647 0.517433
## year2017                          0.25203    0.08923   2.825 0.004735 **
## year2018                          0.10129    0.09229   1.098 0.272407
## year2019                          0.06080    0.09317   0.653 0.514051
## year2025                         -0.24075    0.10068  -2.391 0.016787 *
## water_cap                        -0.69377    0.53338  -1.301 0.193360
## seeded_pool.L:seeding_timespring  0.66755    0.18670   3.575 0.000350 ***
## seeded_pool.Q:seeding_timespring  1.94222    0.23726   8.186 2.70e-16 ***
```

```
## seeded_pool.C:seeding_timespring  1.07629    0.27366   3.933 8.39e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_2,
  revpairwise ~ seeding_time | seeded_pool | year,
```

```r
  type = "response"
  ))
```

```
## $emmeans
## seeded_pool = 6, year = 2015:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            3.336 0.3630 Inf    2.6953     4.130
##  spring          0.612 0.1330 Inf    0.3991     0.937
##
## seeded_pool = 12, year = 2015:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            7.063 0.6180 Inf    5.9502     8.384
##  spring          0.655 0.1400 Inf    0.4313     0.995
##
## seeded_pool = 18, year = 2015:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            8.290 0.7020 Inf    7.0219     9.787
##  spring          0.245 0.0829 Inf    0.1259     0.475
##
## seeded_pool = 33, year = 2015:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall           10.315 0.9030 Inf    8.6898    12.245
##  spring          7.493 0.6620 Inf    6.3012     8.909
##
## seeded_pool = 6, year = 2016:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            3.135 0.3440 Inf    2.5287     3.887
##  spring          0.575 0.1250 Inf    0.3748     0.881
##
## seeded_pool = 12, year = 2016:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            6.637 0.5870 Inf    5.5803     7.895
##  spring          0.616 0.1320 Inf    0.4050     0.936
##
## seeded_pool = 18, year = 2016:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            7.790 0.6680 Inf    6.5847     9.216
##  spring          0.230 0.0780 Inf    0.1182     0.447
##
## seeded_pool = 33, year = 2016:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            9.694 0.8570 Inf    8.1523    11.527
##  spring          7.041 0.6190 Inf    5.9260     8.366
##
## seeded_pool = 6, year = 2017:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            4.293 0.4450 Inf    3.5025     5.261
##  spring          0.787 0.1690 Inf    0.5161     1.200
##
## seeded_pool = 12, year = 2017:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            9.088 0.7370 Inf    7.7523    10.653
##  spring          0.843 0.1780 Inf    0.5577     1.274
##
```

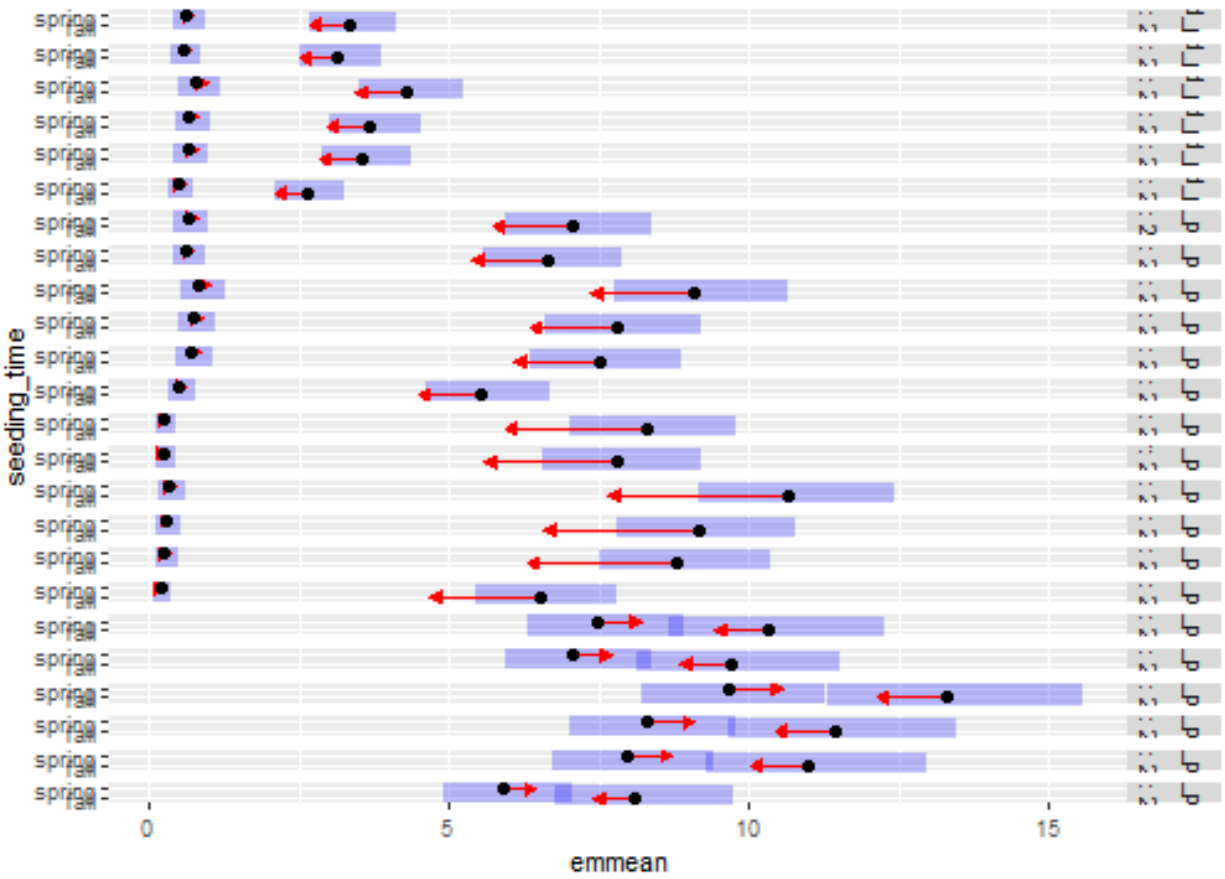
```
## seeded_pool = 18, year = 2017:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall           10.666 0.8330 Inf    9.1523    12.430
##  spring          0.315 0.1060 Inf    0.1624     0.610
##
## seeded_pool = 33, year = 2017:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall           13.272 1.0700 Inf   11.3259    15.553
##  spring          9.640 0.7760 Inf    8.2334    11.288
##
## seeded_pool = 6, year = 2018:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            3.692 0.3930 Inf    2.9969     4.548
##  spring          0.677 0.1460 Inf    0.4427     1.034
##
## seeded_pool = 12, year = 2018:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            7.816 0.6600 Inf    6.6239     9.223
##  spring          0.725 0.1540 Inf    0.4784     1.099
##
## seeded_pool = 18, year = 2018:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            9.173 0.7480 Inf    7.8182    10.764
##  spring          0.271 0.0916 Inf    0.1395     0.525
##
## seeded_pool = 33, year = 2018:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall           11.415 0.9620 Inf    9.6770    13.465
##  spring          8.291 0.6950 Inf    7.0346     9.773
##
## seeded_pool = 6, year = 2019:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            3.545 0.3800 Inf    2.8736     4.374
##  spring          0.650 0.1410 Inf    0.4249     0.994
##
## seeded_pool = 12, year = 2019:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            7.506 0.6410 Inf    6.3490     8.874
##  spring          0.696 0.1480 Inf    0.4591     1.056
##
## seeded_pool = 18, year = 2019:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            8.809 0.7270 Inf    7.4932    10.357
##  spring          0.260 0.0880 Inf    0.1339     0.505
##
## seeded_pool = 33, year = 2019:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall           10.962 0.9340 Inf    9.2754    12.955
##  spring          7.962 0.6750 Inf    6.7426     9.403
##
## seeded_pool = 6, year = 2025:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            2.622 0.2980 Inf    2.0983     3.278
##  spring          0.481 0.1060 Inf    0.3122     0.740
```

```
##
## seeded_pool = 12, year = 2025:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            5.552 0.5190 Inf    4.6219     6.669
##  spring          0.515 0.1110 Inf    0.3373     0.786
##
## seeded_pool = 18, year = 2025:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            6.516 0.5930 Inf    5.4522     7.787
##  spring          0.192 0.0655 Inf    0.0986     0.375
##
## seeded_pool = 33, year = 2025:
##  seeding_time response     SE  df asymp.LCL asymp.UCL
##  fall            8.108 0.7570 Inf    6.7521     9.737
##  spring          5.889 0.5480 Inf    4.9080     7.067
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
## seeded_pool = 6, year = 2015:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.1833 0.04160 Inf    1  -7.477  <.0001
##
## seeded_pool = 12, year = 2015:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0928 0.01980 Inf    1 -11.114  <.0001
##
## seeded_pool = 18, year = 2015:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0295 0.00999 Inf    1 -10.403  <.0001
##
## seeded_pool = 33, year = 2015:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.7263 0.06710 Inf    1  -3.462  0.0005
##
## seeded_pool = 6, year = 2016:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.1833 0.04160 Inf    1  -7.477  <.0001
##
## seeded_pool = 12, year = 2016:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0928 0.01980 Inf    1 -11.114  <.0001
##
## seeded_pool = 18, year = 2016:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0295 0.00999 Inf    1 -10.403  <.0001
##
## seeded_pool = 33, year = 2016:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.7263 0.06710 Inf    1  -3.462  0.0005
##
## seeded_pool = 6, year = 2017:
##  contrast       ratio      SE  df null z.ratio p.value
```

```
##  spring / fall 0.1833 0.04160 Inf    1  -7.477  <.0001
##
## seeded_pool = 12, year = 2017:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0928 0.01980 Inf    1 -11.114  <.0001
##
## seeded_pool = 18, year = 2017:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0295 0.00999 Inf    1 -10.403  <.0001
##
## seeded_pool = 33, year = 2017:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.7263 0.06710 Inf    1  -3.462  0.0005
##
## seeded_pool = 6, year = 2018:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.1833 0.04160 Inf    1  -7.477  <.0001
##
## seeded_pool = 12, year = 2018:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0928 0.01980 Inf    1 -11.114  <.0001
##
## seeded_pool = 18, year = 2018:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0295 0.00999 Inf    1 -10.403  <.0001
##
## seeded_pool = 33, year = 2018:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.7263 0.06710 Inf    1  -3.462  0.0005
##
## seeded_pool = 6, year = 2019:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.1833 0.04160 Inf    1  -7.477  <.0001
##
## seeded_pool = 12, year = 2019:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0928 0.01980 Inf    1 -11.114  <.0001
##
## seeded_pool = 18, year = 2019:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0295 0.00999 Inf    1 -10.403  <.0001
##
## seeded_pool = 33, year = 2019:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.7263 0.06710 Inf    1  -3.462  0.0005
##
## seeded_pool = 6, year = 2025:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.1833 0.04160 Inf    1  -7.477  <.0001
##
## seeded_pool = 12, year = 2025:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0928 0.01980 Inf    1 -11.114  <.0001
##
```

```
## seeded_pool = 18, year = 2025:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.0295 0.00999 Inf    1 -10.403  <.0001
##
## seeded_pool = 33, year = 2025:
##  contrast       ratio      SE  df null z.ratio p.value
##  spring / fall 0.7263 0.06710 Inf    1  -3.462  0.0005
##
## Tests are performed on the log scale
```

```
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
```

```
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         compiler_4.5.1
##  [7] mgcv_1.9-3             vctrs_0.6.5            pkgconfig_2.0.3
## [10] crayon_1.5.3           fastmap_1.2.0          backports_1.5.0
## [13] labeling_0.4.3         utf8_1.2.6             ggstance_0.3.7
## [16] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [19] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [22] later_1.4.2            parallel_4.5.1         R6_2.6.1
## [25] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [28] RColorBrewer_1.1-3     car_3.1-3              boot_1.3-31
## [31] numDeriv_2016.8-1.1    estimability_1.5.1     Rcpp_1.1.0
## [34] iterators_1.0.14       knitr_1.50             zoo_1.8-14
## [37] parameters_0.27.0      httpuv_1.6.16          splines_4.5.1
## [40] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.17.1
## [43] abind_1.4-8            yaml_2.3.10            MuMIn_1.48.11
## [46] doParallel_1.0.17      TMB_1.9.17             codetools_0.2-20
## [49] lattice_0.22-7         plyr_1.8.9             shiny_1.11.1
## [52] withr_3.0.2            bayestestR_0.16.1      coda_0.19-4.1
## [55] evaluate_1.0.4         marginaleffects_0.28.0 pillar_1.11.0
## [58] gap_1.6                carData_3.0-5          foreach_1.5.2
## [61] stats4_4.5.1           reformulas_0.4.1       insight_1.3.1
## [64] generics_0.1.4         vroom_1.6.5            rprojroot_2.1.1
## [67] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [70] xtable_1.8-4           glue_1.8.0             tools_4.5.1
## [73] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.1
## [76] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [79] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [82] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [85] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [88] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [91] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```